

Simulating Wildfire Mitigation: A Multi-Agent Approach to Community Behavior

Anonymous submission

Abstract

Homeowners vary widely in their attitudes, financial constraints, and risk exposure, making wildfire mitigation responses difficult to study at scale. We present an LLM-based multi-agent simulation grounded in twenty qualitative interviews and investigate whether LLM-as-judge protocols can reliably evaluate persona fidelity. Agents are instantiated from interview-derived and synthetic personas and interact over a 60-day wildfire mitigation intervention.

We compare two evaluation paradigms: pointwise scoring, which independently assesses consistency with a target persona, and forced-choice attribution, which identifies the most likely author among all candidate personas. Pointwise scoring provides only moderate discrimination ($AUC = 0.72$ among interview-grounded personas; $AUC = 0.84$ with a synthetic persona included), largely capturing coarse categorical differences. In contrast, forced-choice attribution accurately identifies persona authorship, achieving 85.8% accuracy among interview-grounded personas (25% chance) and 88.7% when the synthetic persona is included (20% chance). These results indicate that the apparent weakness lies in the evaluation protocol rather than the persona construction.

We also find that attribution accuracy is largely unchanged when conversational memory is removed (85.8% vs. 81.7%), though a moderate memory effect cannot be ruled out given the sample size. This result suggests that interview-derived persona specifications, rather than accumulated interaction history, are responsible for most of the identity signal over simulations of this duration. To summarize, our findings suggest that empirically grounded LLM-agent simulations can produce behavior that is both distinguishable and quantitatively evaluable from small qualitative datasets, providing a scalable framework for: (1) studying human responses to wildfire risk and (2) validating persona fidelity in LLM-based social simulations.

Introduction

In recent years, wildfire destruction has reached catastrophic levels; California alone lost over 41,000 structures between 2018 and 2023 (Wilkin et al. 2025). Despite stricter defensible space mandates under California’s AB 3074 and SB 504

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and increased agency outreach, a wildfire mitigation compliance gap persists. Homeowners often fail to take even cursory steps toward fire-resilience, leaving communities vulnerable to escalating threats (Wilkin et al. 2025; Semancik 2025). Understanding the behavioral and social factors deterring homeowners from taking action would allow fire officials, policymakers, and insurers to identify effective interventions. Generative AI is well-positioned to contribute here: LLM agents can be grounded in data about homeowner populations to simulate psychologically realistic behavior across many agents and intervention sequences, enabling rapid, iterative testing of how various factors influence behavior (Park et al. 2023).

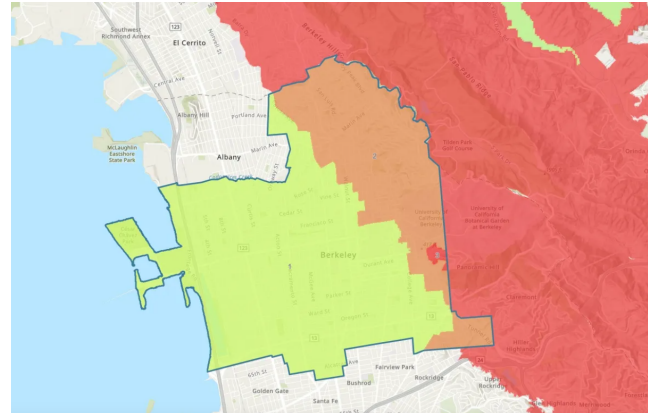


Figure 1: Berkeley Fire Zones. Zone 1 (yellow, flats) has the lowest risk. Zone 2 (orange, Berkeley Hills) has higher risk due to building density and wildland proximity. Zone 3 (red, Panoramic Way) has the highest risk due to evacuation and access challenges. Image Source: Gecan (2025).

Our system leverages homeowner interview transcripts and an intervention schedule as input, and uses a retrieval-augmented agentic workflow with reflection to produce per-tick decisions, reasoning, and evolving belief states for simulated homeowners.

The sections that follow describe our data pipeline, system architecture, simulation experiments, and findings.

Related Work

Research with wildland–urban interface (WUI) homeowners finds that individuals’ wildfire mitigation efforts are informed by numerous factors including perceived risk, perceived effectiveness of mitigation activities, home attachment, prior experiences with wildfires, and trust in agencies (Ghasemi et al. 2020). We leverage qualitative interviews to characterize relevant behavioral and attitudinal dimensions of homeowners represented in our simulation. Social science literature also documents that whether or not a policy is mandatory, neighborhood norms, and insurance considerations shape defensible-space compliance (Winter et al. 2009). These insights inform the interventions that agents in our simulation respond to: ordinance information, inspections, social pressure, and communication regarding insurance status.

Our simulation architecture draws directly from Park et al. (2023), which established the core design patterns we adopt: a retrieval mechanism scoring memories by recency, importance, and relevance; a two-step reflection process; and an append-only memory stream as the foundation for agent cognition (Park et al. 2023). We extend this architecture by grounding agents in qualitative data from semi-structured interviews with homeowners, focusing on behavioral and attitudinal dimensions relevant to this domain, rather than researcher-authored fictional personas. This grounding is motivated by evidence that appropriately conditioned LLMs can emulate the attitudes of demographically distinct sub-populations (Argyle et al. 2023).

Agent-based models, such as ours, are useful for exploring uncertain circumstances, as they allow investigation of interaction between various inputs, outcomes, and adaptive policies (Lempert 2002). Perhaps closest to our approach, Hou et al. (2025) use generative agents grounded in census data to inform public-health policy, validating persona fidelity – how consistent and reasonable agents’ responses are – with pointwise LLM-judge scoring, wherein the judge rates each response against its target persona on a 1–5 scale (Hou et al. 2025). We interrogate a similar pointwise protocol.

We leverage LLM-as-a-judge to evaluate agent responses, as it is increasingly regarded as a scalable and explainable substitute for human rating. Strong judges approximate human preference; Zheng et al. (2023) reported over 80% judge–human agreement, comparable to human–human agreement (Zheng et al. 2023).

Despite their utility, LLM judges are imperfect. For instance, they exhibit systematic biases based on position, verbosity, and self-preference: preference for responses generated by the same model used for judging (Gu et al. 2025). Further, their accuracy is influenced by both prompt construction and scoring mechanisms. For example, comparative judging formats can be more robust than pointwise scoring, as comparison may better surface granular distinctions (Gu et al. 2025). In response to concerns about LLM judges, we compare a pointwise judge protocol against a forced-choice attribution format, and use a persona-swap diagnostic and a synthetic-renter control to evaluate our judge’s ability to assess agents’ behavior and attitudes.

Our work offers a framework for conducting a low-cost simulation for studying homeowner responses to wildfire-

mitigation interventions to inform policy. It further offers insight into how accurately LLM-as-judge protocols can measure persona distinctions in social simulations, to inform future work.

Data Sources and Inputs

We leverage 20 semi-structured interviews with homeowners in the Berkeley Hills, CA, conducted with English-speaking members of the Firewise Community, a voluntary program to reduce neighborhood wildfire risk. The sample was gender-balanced, predominantly composed of seniors, and diverse in education, financial status, and political ideology. Eleven interviews were transcribed using Google Pixel’s on-device Gemini Nano model (57,118 words total, avg. 5,100/transcript); the remaining nine were documented by note-takers.

A pre-processing pipeline reconciled the granularity gap between verbatim AI transcripts and summarized notes, standardized formatting, and anonymized interviewees. Using Claude Sonnet 4.5, we extracted structured personas (compliance status, insurance status, risk classification) via a three-stage protocol:

- **Held-out Responses:** key reflections quarantined per intervention during tuning, to avoid over-reliance on single quotes.
- **Memory Seeds:** episodic contexts and actions from the transcript, weighted by importance (1–10).
- **Seed Narrative:** the agent’s stable persona and voice.

Outputs were saved as YAML and cross-checked against transcripts by two team members. To evaluate cross-model generalization, we tested the pipeline using Gemini 2.5 Pro. While Gemini’s extractions were accurate, they lacked coverage, omitting several discrete memory seeds and fine-grained details. This confirmed the structural robustness of the protocol but justified retaining Claude Sonnet 4.5 to maximize behavioral depth.

To address the inherent bias of the “compliant” Firewise sample, we created synthetic agents, built from proxy data of less-engaged neighbors, for stress-testing retrieval. In addition, a synthetic agent was included in the simulation to represent a diverse set of homeowner perspectives. While a single synthetic agent is insufficient to represent the full diversity of the broader community, it serves as a necessary proof-of-concept to test whether the system’s cognitive architecture can generalize to uncooperative behavioral profiles.

Interventions. To ground the simulation in real-world stressors, we developed a 60-day intervention schedule (`baseline.yaml`; Figure 2): Zone 0 notifications under the Berkeley Ember Ordinance City of Berkeley (2025), a fire department inspection, neighbor engagement, a Firewise campaign, an insurance non-renewal notice, and mitigation resources via the Resident Assistance Program (RAP).

System Usage Example

Our system provides the first step to creating a tool for community officials and policy makers. To ground its utility, take, for instance, a Berkeley city official. This city official might

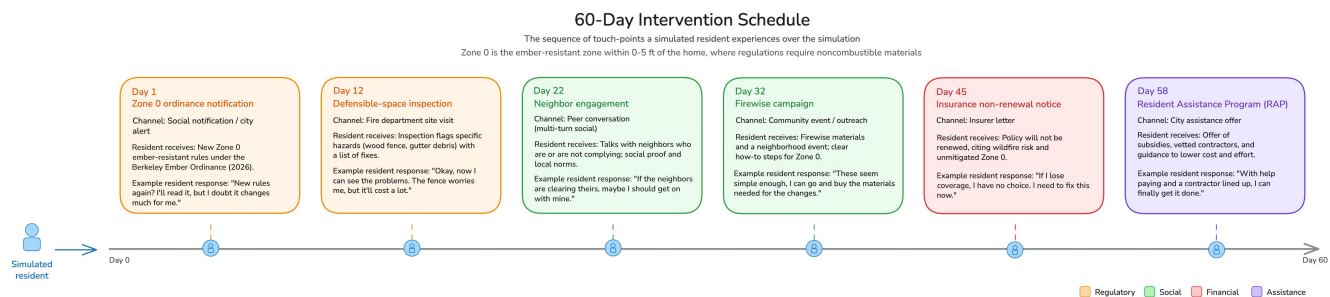


Figure 2: The 60-day intervention schedule, spanning ordinance notification, inspection, social pressure, community outreach, insurance non-renewal, and resource assistance.

use this system to design a more effective and community-focused outreach campaign before it reaches real residents. See Figure 3 for a visual of how the system would work. The official would start by providing two inputs. First, a set of homeowner profiles, each capturing a real resident’s attitudes toward fire risk, their history with insurance and inspections, their financial situation, and their level of trust in city agencies. Second, a proposed intervention schedule: a sequence of interventions the city plans to enact over coming months, such as written ordinance notices, fire department defensible space inspections, community events, and offers of assistance.

The simulation then runs through those sixty days, simulating each homeowner. The output is a detailed log of every homeowner’s decisions and reasoning at each touchpoint. The official can read these traces in plain language: “Laura said she would prioritize removing the wood fence after the inspection, but by day 48 she had not followed through and cited cost as the barrier.” They can compare how residents with different characteristics react and respond to interventions and identify where a different approach (i.e. a phone call rather than a formal notice) might produce a better compliance outcome, or the same compliance outcome with less burden on residents. This could result in a campaign that has been stress-tested with the community it serves, to produce better outcomes for both officials and residents.

System Architecture

Architecture Overview

The system is organized into seven layers: Inputs, Config, Engine, Environment, Agent Cognition, Output, and Evaluation. The per-tick flow proceeds as follows: the scheduler pulls due events from a heapq priority queue and returns them to the engine; the engine routes each event through the appropriate channel framing in `channels.py`; each targeted agent runs the full perceive, retrieve, decide, act, store cognition cycle; the reflection check fires if the cumulative importance threshold is exceeded; and the logger writes all outputs to JSONL.

The Inputs Layer supplies three files at run time: `agents.yaml` (agent personas, memory seeds, and seed narratives), `baseline.yaml` (the intervention schedule), and the config YAMLS (model and hyperparameter settings). The Config Layer parses these into three config objects that feed the rest

of the system. The Engine Layer (`simulation.py`, `scheduler.py`, `run_simulation.ipynb`) orchestrates the tick loop. The Environment Layer (`channels.py`, `network.py`) wraps raw events in channel-specific framing, such as official mail or direct experiences shaping which memories the agent retrieves when processing each event. The Agent Cognition Layer runs the perceive-retrieve-decide-act-store cycle per agent per tick; `conversation.py` handles multi-turn social interactions that interrupt the tick cycle when agents interact directly. All LLM calls are routed through the LLM Layer (`client.py`), which dispatches decisions and reflections to the configured language model and embeddings to a local sentence-transformer. The Output Layer logs all decisions, memories, reflections, and conversations to JSONL via `logger.py`. The Evaluation Layer (`run_evaluation.ipynb`) reads those outputs, runs the LLM judge and exports scored results to an Excel workbook for review.

Agent Cognition Deep Dive

Memory stream: Each agent maintains an append-only list of typed Memory objects carrying a timestamp, natural language description, importance score (LLM-assigned, 1–10), vector embedding, memory type (observation, decision, reflection, or conversation), and a `last_accessed` field. Recency decay is calculated from `last_accessed` rather than creation time, so frequently retrieved memories remain fresh even if old.

Retrieval: `retrieval.py` scores every memory in the stream as a weighted combination of recency, importance, and relevance, each normalized to [0,1] before weighting, following (Park et al. 2023). The top-K highest-scoring memories are returned and assembled into the agent’s prompt context.

Reflection: After each tick, `reflection.py` checks whether the cumulative importance of unprocessed memories exceeds a configurable threshold. If so, it executes a two-step process: first generating N high-level questions from recent memories, then retrieving relevant memories per question and synthesizing a first-person insight with citations to source memories. Insights are stored back to the memory stream as high-importance reflection memories.

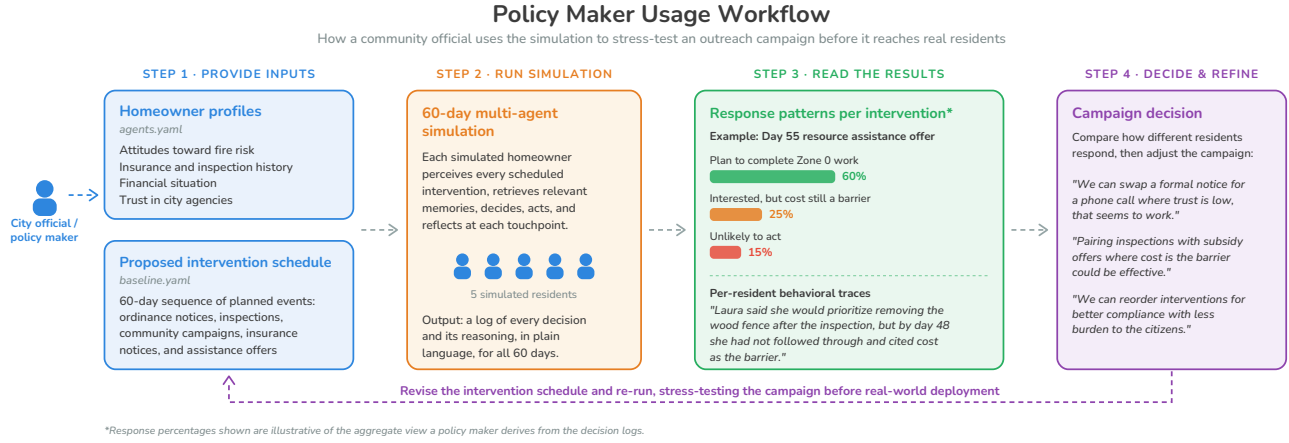


Figure 3: Policy maker usage workflow: a city official provides homeowner profiles and a proposed intervention schedule, runs the 60-day simulation, reads aggregate response patterns and per-resident behavioral traces, and refines the campaign before real-world deployment.

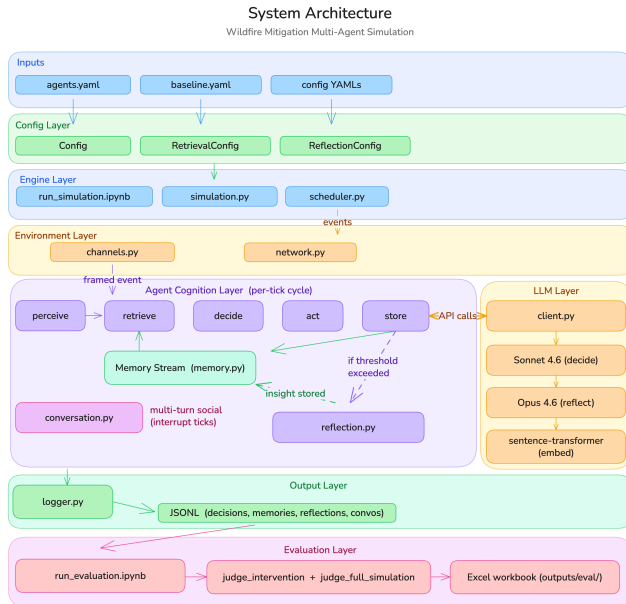


Figure 4: The seven-layer system architecture.

Experiments and Evaluation

Agent Tuning and Evaluation

Evaluation Approach Agent behavior was evaluated by testing each component of the agent cognition pipeline independently prior to running the full simulation. Agent Margaret was used for agent tuning, as she was modeled off a Berkeley Hills homeowner who had ground truth responses for all six intervention types present in the simulation: insurance non-renewal, defensible space inspection, neighbor pressure, Firewise outreach, and resource assistance. Ground truth responses made it possible to establish held-out responses for every intervention, allowing for end-to-end validation of the pipeline. Other agents did not have held-out

responses for every intervention, reflecting the reality that interviewees have not directly experienced each simulated scenario. This simulation is useful as a tool to explore how individuals will respond to novel mitigation efforts. All LLM-as-judge evaluations used Claude Sonnet (claude-sonnet-4-6) at temperature=0.

Memory Seeding Evaluation Prior to testing retrieval and generation, we evaluated whether the agent’s seed memories provide sufficient background context to respond in character across all six interventions. The judge assessed whether the seeds establish the right attitudes, experiences, and knowledge to enable an authentic response. The judge was not shown the held out responses because that would have compromised validity. It checks whether the memory seeds are detailed enough for the agent to respond like the real person would for every scenario. For each scenario, the judge output an adequacy score (1–10), a list of context missing from memories, and a one to two sentence explanation of its scoring.

Memory Retrieval Hyperparameter Tuning and Evaluation

At decision time, the retrieval system selects which memories to pass to the agent by scoring every memory in the stream using a weighted combination of three dimensions: recency (α), importance (β), and semantic relevance (γ). The top-K memories by this score are included in the agent’s prompt. We tested nine retrieval configurations across all six interventions, varying top-K (5, 8, 12), the three dimension weights, and retrieval mode (dense, sparse, hybrid). For each configuration, the LLM judge scored whether the retrieved memories were relevant for that scenario (1–10), identified any important memories that were missed, and provided a one to two sentence critique. Highest performing configurations were identified per scenario. A single configuration was selected to be used across all scenarios in the simulation, balancing performance across intervention types: top-K = 8, $\alpha = 1.0$, $\beta = 1.0$, $\gamma = 1.0$, mode = dense.

Reflection Hyperparameter Tuning and Evaluation Reflections are higher-level beliefs or insights that an agent synthesizes from patterns across multiple memories, stored back into the memory stream as a new memory of type reflection. Following Park et al.’s (2023) methodology, reflection was triggered when the cumulative importance of unprocessed memories (those added since the last reflection) exceeded a threshold (Park et al. 2023). We tuned three hyperparameters: threshold (50, 100, 150), number of questions (3, 5), and reflection importance score (fixed at 8). The number of questions controlled how many introspective questions the model generated in the first step of reflection, each of which produced one new reflection memory. Reflection importance was fixed at 8 so that reflections surfaced consistently in future decision prompts without dominating the agent’s observational memories. Agent Margaret’s 55 seed memories had a cumulative importance of approximately 363, so even a threshold of 150 triggered reflection immediately at the seed stage. Park et al. (2023) used 150 as a threshold, and we included lower thresholds of 50 and 100 since our simulation operates at a smaller scale and has fewer events overall (Park et al. 2023). The judge scores each configuration on insight quality and accuracy of the synthesized beliefs against the source memories, both on a 1–10 scale. The best configuration found here was: threshold=50, number of questions=3, importance=8.

Simulation Evaluation

The simulation lasted 60 days and included five agents—Laura, Linda, Walter, Margaret, and Miriam Voss—across six interventions: a Firewise ordinance notification (day 1), a fire department defensible-space inspection (day 12), social pressure (day 24), a community Firewise campaign (day 36), an insurance non-renewal notice (day 48), and a resource-assistance email (day 55). Laura, Linda, Walter, and Margaret were derived from homeowner interviews to span a range of compliance statuses, financial situations, and institutional trust levels among Berkeley Hills residents. Miriam Voss was created synthetically as a non-compliant renter, to represent a perspective not captured in the interviews.

The simulation was run under four conditions (Table 1) to test the influence of cognitive architecture parameters and model selection. All runs used the locked retrieval config (top- K =8, equal weights, dense) and reflection config (threshold=50, 3 questions, importance=8) fixed during agent tuning.

Evaluation Design We examine whether the pointwise LLM-as-a-judge protocol, used in prior instances to validate persona agents, actually measures persona fidelity (Hou et al. 2025). Early simulation runs reported near-ceiling pointwise judge scores (behavioral plausibility, persona consistency, and intervention responsiveness all clustering between 4.8 and 5.0 out of 5), which might be read as evidence that the agents were persona-consistent. However, further investigation revealed that this inference does not hold: an LLM judge conducting pointwise persona-consistency scoring assigned an agent’s response a similar score whether it was evaluated against its true author’s seed or a different agent’s seed. We

Condition	Description
Baseline	Claude Sonnet 4.6 for decisions and reflection; full memory stream, retrieval, and reflection
Ablation 1 (No Reflection)	Claude Sonnet 4.6; memory stream and retrieval active, reflection disabled
Ablation 2 (No Memory or Reflection)	Claude Sonnet 4.6; seed personality only, no memory or retrieval
Budget	Claude Haiku 4.5 for decisions and reflection; full memory stream, retrieval, and reflection

Table 1: The four simulation conditions.

therefore treat the evaluation instrument as an object of study, examining pointwise scoring against a persona-swap control and forced-choice attribution.

Judge configuration. All simulation-evaluation judgments use `openai/gpt-5.4` at temperature 0 under a locked rubric. The judge is deliberately drawn from a different model family than the agents, which run on `claude-sonnet-4-6`, as scoring a model family with itself risks self-preference bias (Gu et al. 2025). The unit of analysis is one agent $\times$ one intervention, rather than a whole trajectory, in which the memory-dependent interventions would be averaged together with memory-independent ones, diluting their signal. Each judgment scores a single dimension per call with inputs restricted by construction: behavioral plausibility (BP) and intervention responsiveness (IR) see only the agent’s objective circumstances (e.g., property type, risk zone, compliance status, insurance status) and not its seed narrative, while persona consistency (PC) sees the seed narrative and memory seeds but not the circumstances. The rubric anchors all five scale points and requires integer scores.

Replication. The judge is non-deterministic even at temperature 0: across five identical passes, the same item receives a different score roughly 27% of the time, enough to move summary statistics across effect-size boundaries (single-pass AUC ranged 0.69–0.78). We therefore report means over five passes for each measure (five-pass mean AUC 0.724, s.d. $\approx$ 0.03).

Protocols. Pointwise scoring rates one response against one target persona (1–5). The swap control scores each response against its own seed and every other agent’s seed, comparing matched against mismatched scores. Forced-choice attribution shows the judge one response and all five candidate seeds and asks it to name the author (chance 25% for owners, 20% with the renter).

Pointwise Scoring Does Not Separate Conditions We scored every agent decision across all four conditions with the pointwise protocol, averaging five repeated passes (1,800 total judgments). Scores compressed toward the top of the

Table 2: Pointwise scores across conditions (1,800 judgments, 5 agents $\times$ 5 passes). Differences are negligible ($|\Delta_{\text{Mean}}| \leq 0.02$ vs. baseline, within noise SD 0.02–0.06), except for a ≈ 0.10 persona-consistency drop in Ablation 2 and Budget.

Condition	BP	PC	IR	Mean
Baseline	4.41	4.48	4.69	4.53
Ablation 1	4.42	4.47	4.73	4.54
Ablation 2	4.49	4.38	4.77	4.54
Budget	4.40	4.37	4.77	4.51

scale: 99.3% were a 4 or 5, and condition-level means spanned only 4.51 to 4.55 (Table 2). This inter-condition gap matches the variation between repeated passes of the same condition (SD 0.02–0.06). Furthermore, the full ablation (no memory or reflection) was statistically indistinguishable from Baseline (paired mean difference 0.02, within-pair SD 0.35). Under pointwise scoring, removing the entire cognitive architecture produces no measurable overall difference. The only directional signal was persona consistency, where the no-memory ablation and Haiku configuration each scored ~ 0.10 below Baseline (4.38 and 4.37 vs. 4.48).

While repeated passes rule out judge randomness, near-ceiling scores leave ambiguity: either the cognitive architecture genuinely contributes little over a 60-day horizon, or pointwise scoring lacks discriminative power. This ambiguity motivates a protocol requiring the judge to discriminate.

The Swap Control: Pointwise Scoring Does Not Separate Personas The persona-swap control isolates the instrument to test this discriminative power. Restricted to the four interview-derived owners, a response scored against its true author’s seed averages 4.33/5. Scored against a different owner’s seed, the same responses drop only modestly to 3.69/5—a separation of +0.64 (Cliff’s $\delta = 0.448$, 95% CI [0.370, 0.521], medium; discrimination AUC 0.724).

This limited sensitivity is confined to demographically similar owners. Swaps crossing the renter/owner boundary are consistently identified, raising the AUC to 0.839 once the renter (Miriam) is included. Consequently, high pointwise scores do not establish true persona consistency, as responses score nearly as well against another owner’s seed. Whether this reflects a limitation of the judge or genuine behavioral similarity among owners is addressed in the discussion.

Forced-Choice Attribution Recovers Distinctness The failures above are failures of the pointwise *protocol*, not evidence that the agents are indistinguishable. Shown one response alongside all five candidate seeds and asked to name the author, the judge identifies the true owner 85.8% of the time against a 25% chance baseline (Wilson 95% CI [78.5%, 91.0%]; binomial $p < 0.001$); including the renter, the rate is 88.7% against 20% chance, and Miriam is attributed correctly in 100% of instances (Table 3). Forced-choice attribution thus shows that the agents are distinct from one another; it does not establish fidelity to the real homeowners they are drawn from, which would require held-out human validation we do not have.

Table 3: Forced-choice attribution and swap-control discrimination (gpt-5.4, $T = 0$, 5-pass means). Owner attribution is a 4-way choice (25% chance); renter Miriam serves as a positive control. Attribution persists without memory/reflection (Ablation 2) and under a smaller model (Budget).

Measure	Owners	With renter
Swap-control AUC	0.724	0.839
Attribution — Baseline	85.8%	88.7%
Attribution — Ablation 2	81.7%	—
Attribution — Budget (Haiku)	91.7%	—
Miriam (positive control)	100%	—
Chance baseline	25%	20%

Using attribution as the measurement, we ask whether the memory and reflection machinery makes agents more attributable. Owner attribution is 85.8% under Baseline and 81.7% under Ablation 2 (seed personality only), a difference of -4.2 percentage points that is not statistically detectable: McNemar’s exact test on the 120 paired judgments gives $p = 0.458$, and the bootstrap 95% CI on the paired difference is $[-14.2\%, +6.7\%]$. Because that interval is wide, we read this as no *detectable* difference rather than equivalence; the comparison is underpowered at five agents. A pre-registered check for a memory-dependent interaction (a larger drop on the memory-dependent days 24, 48, 55) is likewise non-significant (Mann–Whitney $p = 0.103$). The identity signal that attribution recovers therefore appears to be carried by the seed rather than by accumulated memory.

Cost and Latency The architecture’s cost is dominated by context size: Baseline prompt context is roughly double that of Ablation 2 ($\sim 180,000$ vs. $\sim 66,000$ tokens), and the Budget condition (Haiku) reduces agent cost about threefold without a proportional latency reduction (Table 4). Consistent with the identity signal being seed-driven, this cost reduction does not appear to degrade it: owner attribution under Budget is 91.7%, comparable to Baseline’s 85.8% and far above chance. A smaller model is therefore a reasonable default for proof-of-concept deployment.

Quadratic-Weighted Cohen’s Kappa ($n = 15$) Human raters agreed only fairly on Behavioral Plausibility ($\kappa=0.250$) and Intervention Responsiveness ($\kappa=0.286$), and only slightly on Persona Consistency ($\kappa=0.182$). Human-LLM agreement was moderate for Persona Consistency ($\kappa=0.412$) but near-zero for Intervention Responsiveness ($\kappa=0.100$) and negative for Behavioral Plausibility ($\kappa=-0.313$), falling below chance-level agreement. These modest kappa values coexist with high raw percent agreement (60–97%) and likely reflect a ceiling effect: with $n = 15$ and most agent responses rated 4 or 5, the skewed score distribution inflates chance-agreement, compressing the achievable kappa regardless of underlying rater consistency.

Discussion

We evaluated the interview-grounded agents with two judging protocols, which gave different pictures of the same

Condition	Quality (1–5)	Agent Cost / Run (USD)	Latency / Run (s)	Total Cost / Run (USD)
Baseline (Claude Sonnet 4.6)	4.53	\$0.78	605	\$1.51
Budget (Claude Haiku 4.5)	4.51	\$0.26	500	\$0.99

Table 4: Cost and latency per run. Quality is the independent per-intervention judge score (GPT-5.4, five passes averaged): mean of BP, PC, IR across agents and interventions. Agent cost and latency cover the simulation’s decision and reflection calls only.

Comparison	BP	PC	IR
Human-Human	0.250	0.182	0.286
Human-LLM	-0.313	0.412	0.100

Table 5: Quadratic-Weighted Cohen’s Kappa for Human-Human and Human-LLM Evaluation. For Human-LLM evaluation, human scores were averaged across the two raters.

agents. Attribution identified authors well above chance, showing the agents are distinct, so the weak separation under pointwise scoring is a product of the protocol rather than the agents, which scored all four conditions near-identically (4.51–4.54) despite spanning full memory, ablated memory, and a smaller model. We cannot resolve whether this pointwise weakness stems from limited judge sensitivity to granular persona differences or from genuine similarity among four owners who are demographically alike and uniformly compliant. More generally, high pointwise persona-consistency scores should not be read as evidence of persona fidelity without a discriminative check such as attribution. This advantage is specific to identity tasks, where the judge need only rank the true persona first; for rating response quality, absolute scoring is often more robust, as pairwise judgments are more easily manipulated by spurious features (Tripathi et al. 2025).

The agents’ distinctness appears to come from the persona seed rather than accumulated memory: attribution accuracy was essentially unchanged with memory and reflection removed (85.8% Baseline vs. 81.7% Ablation 2, not statistically detectable). This is consistent with the richness of the seeds—in contrast to the brief fictional profiles used by Park et al. (2023), our multi-paragraph interview-grounded seeds encode enough personality and behavioral detail for agents to remain distinguishable without retrieval across most interventions. A moderate memory effect cannot be excluded at this sample size, but over simulations of this length the identity signal appears seed-driven rather than memory-driven. This also makes the simulation practical to run at scale: the Budget configuration (Haiku) ran at roughly one third the agent cost of the full configuration (\$0.26 vs. \$0.78/run) at comparable latency (500s vs. 605s), without degrading the identity signal—owner attribution was 91.7%, comparable to Baseline’s 85.8% and far above the 25% chance baseline. Together, these results indicate the identity signal is carried by the seed and is robust to both removing the memory architecture and substituting a smaller model, a reasonable default for proof-of-concept deployment.

Three further points are worth noting. First, the synthetic

reenter Miriam functions as a categorical positive control: attributed with perfect accuracy, her categorical difference from the owners is enough for a seed-blinded judge to separate her, raising discrimination when she is included (owners-only AUC 0.72 to 0.84). Second, the judge is non-deterministic even at fixed temperature: roughly a quarter of judgments change across identical passes (26.5% pairwise flip rate), enough to shift summary statistics across effect-size boundaries—single-pass measures near such a boundary should be treated as unstable and reported as multi-pass means. Third, this is corroborated by human-LLM agreement, weakest on Behavioral Plausibility and Intervention Responsiveness ($\kappa = -0.313, 0.100$) and moderate only for Persona Consistency ($\kappa = 0.412$)—the dimension carrying most of the attribution signal. Overall, agents built from a small set of interviews were distinct enough to tell apart and could be scored quantitatively, at low and largely model-agnostic cost—suggesting that studying homeowner responses to wildfire interventions is feasible without large datasets or high spend, provided attribution rather than pointwise scoring is used to check distinctness.

Conclusions and Future Work

This work extends Park et al. (2023)’s generative agent architecture to policy-relevant behavior using empirically derived personas. Interview-grounded seeds produced high-fidelity agent behavior independent of memory retrieval within single interventions, unlike prior work using sparse fictional seeds; memory may matter more over longer horizons, once accumulated experience diverges from seed content. Human raters and the LLM judge diverged most on Behavioral Plausibility and Intervention Responsiveness, with Persona Consistency showing the strongest—though still only moderate—agreement, suggesting the former two dimensions warrant closer prompt calibration.

Primary limitations include a small, homogeneous sample ($n = 5$); evaluation against seed personality rather than held-out ground truth; score compression limiting Kappa estimates; and a 60-day horizon too short to fully surface memory-retrieval effects. Future work should prioritize longer runs to surface memory-architecture effects, sensitivity analysis across intervention orderings and population compositions, held-out validation against real interview responses, and a larger, more diverse agent population.

This work demonstrates that interview-grounded multi-agent simulation can produce persona-consistent homeowner responses at low cost, offering a practical tool for wildfire-intervention planning.

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